Question ID: d675744f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

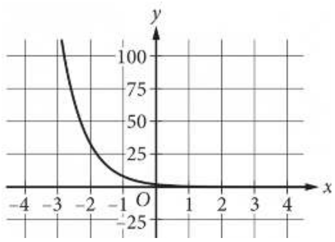
Question

$y = 4(2^x)$

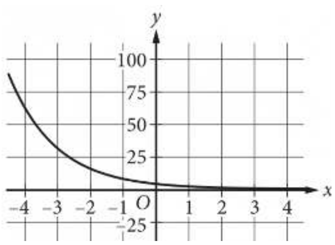
Which of the following is the graph in the xy-plane of the given equation?

Answer

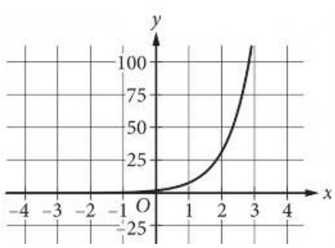
A.



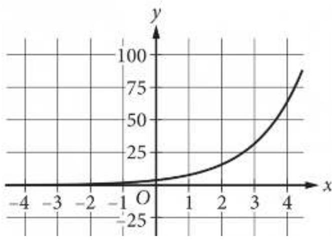
B.



C.



D.



Correct Answer: D

Rationale

Choice D is correct. The y-intercept of the graph of an equation is the point $(0,b)$, where b is the value of y when $x=0$. For the given equation, $y=4$ when $x=0$. It follows that the y-intercept of the graph of the given equation is $(0,4)$. Additionally, for the given equation, the value of y doubles for each increase of 1 in the value of x. Therefore, the graph contains the points $(1,8)$, $(2,16)$, $(3,32)$, and $(4,64)$. Only the graph shown in choice D passes through these points.

Choices A and B are incorrect because these are graphs of decreasing, not increasing, exponential functions. Choice C is incorrect because the value of y increases by a growth factor greater than 2 for each increase of 1 in the value of x.